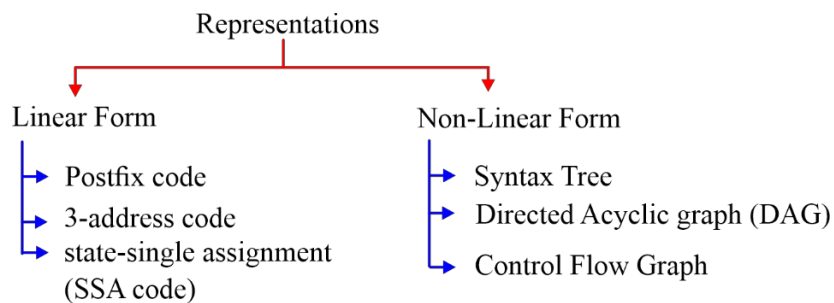


# 5

# INTERMEDIATE, CODE OPTIMIZATION

## 5.1 Introduction



### Example Expression:

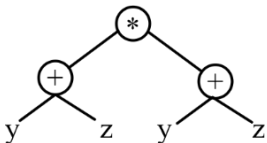
$(y + z) * (y + z)$

Post fix  $\rightarrow$   $yz + yz + *$

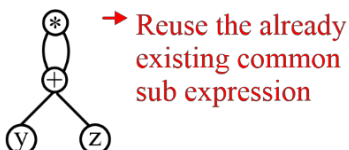
SSA  $\rightarrow$   $t_1 = y + z$   
 $t_2 = t_1 * t_2$   $\Rightarrow f_1$  and  $f_2$  cannot be reassigned

3AC  $\rightarrow$   $t_1 = y + z$   
 $t_2 = t_1 \times t_2$

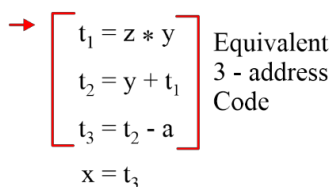
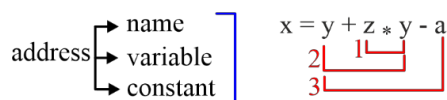
Syntax Tree  $\rightarrow$



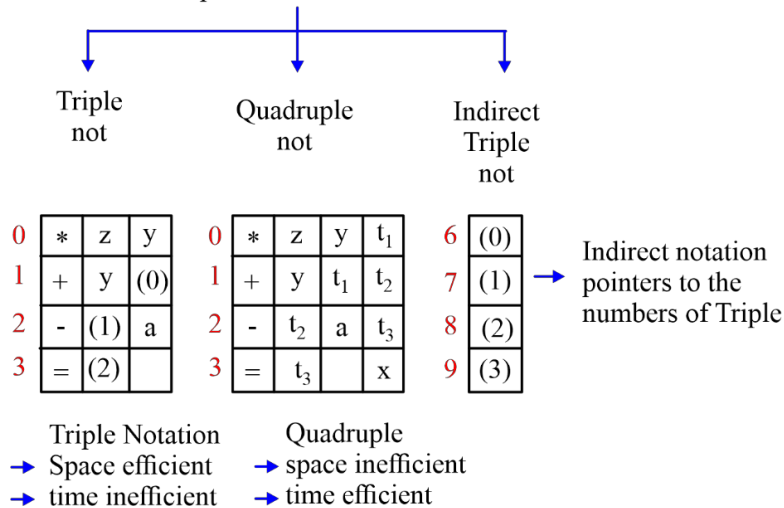
DAG  $\rightarrow$



**3-address Code:** Code in which, at most 3 addresses. [including LHS]

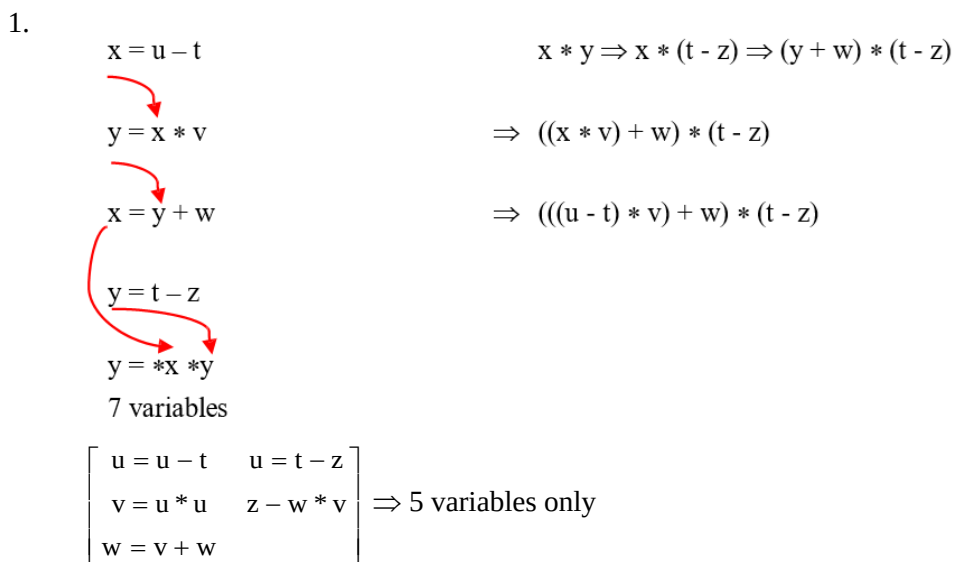


Representation of 3AC



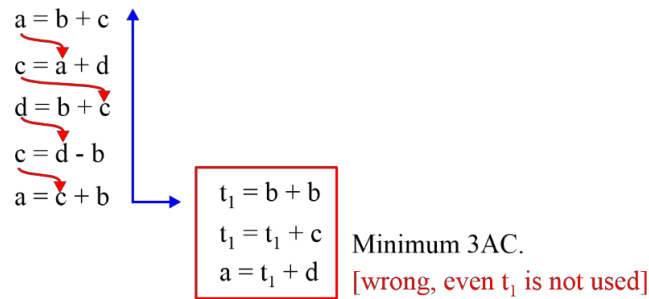
- 3AC done using operator precedence.

**Find minimum number of variables required in equivalent 3AC:**



⇒ Evaluating the expression:

$$\begin{aligned} a &\Rightarrow c + b \Rightarrow d - b + b \Rightarrow b + c - b + b \\ &\Rightarrow b + a + d - b + b \\ &\Rightarrow b + b + c + d - b + b \\ &\Rightarrow b + b + c + d \end{aligned}$$



∴ Minimum:

$$\left\{ \begin{array}{l} b = b + b \\ c = b + c \\ a = c + d \end{array} \right\} \Rightarrow \text{only 3 variables [most optimal]}$$

### 5.1.1 Static Single Assignment Code: (SSA Code)

Every variable (address) in the code has single assignment [single meaning] + 3AC.

1.  $x = u - t$   
 $y = x * u$   
 $x = y + w$   
 $y = t - z$   
 $y = x * y$

Find SSA?

⇒  $[u, t, v, w, z]$  are already assigned so we can't use them.

Equivalent SSA Code:

$x = u - t$   $y = x * u$

$p = y + w$

$q = t - x$

$r = p * q$

in use:  $x, y, p, q, r \Rightarrow$  additional

∴ Total Variable ⇒ 10.

2.  $p = a - b$

$q = p * c$

$p = u * v$

$q = p + q$

⇒  $[a, b, c, u, v]$  are already assigned,

Equivalent SSA Code:

$p = a - b$

$q = p * c$

$p_1 = v * u$

$q_2 = p_1 + q$

in use:  $p, q, p_1, q_2$

∴ Total Variable = 9